Player Agency and Content Retention in Educational Games

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Introduction

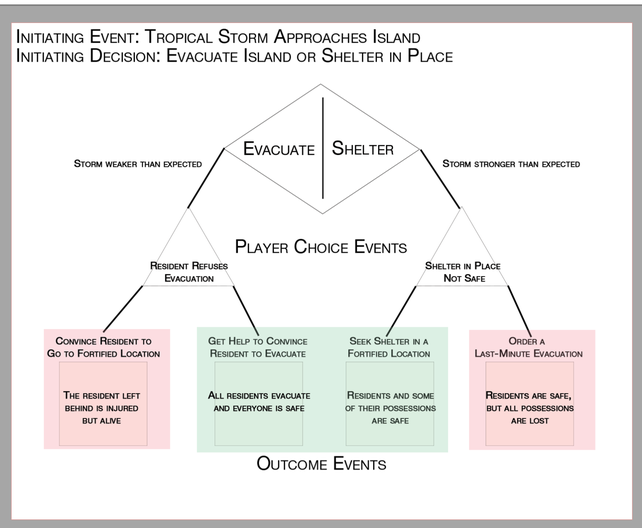
Context and Significance

There are an estimated 155 million video game players in the US as of 2015, a full 23% of whom are aged 18 or younger[1]. The average gamer aged 13 and up spends 6.3 hours a week gaming (the length of a typical school day)[2]. Gaming is now prevalent enough in mainstream culture that it is an inevitable inclusion in the classroom as we seek to engage students and improve learning outcomes. In order to create effective educational games, we need to understand how game design elements impact aspects of learning.

Player Agency, best defined as "the ability to change the course of one's experience"[3], is an important game design element because it strongly influences a player's sense of immersion in the game environment and early research suggests that this immersion has positive implications for learning in educational games[4]. Player agency may also influence motivation and attention of players if it increases the perceived gameness of the game.

Player agency is further subdivided into theoretical (objective) agency, which is one's objective capacity for influencing the course of events in the game, and perceptual (subjective) agency, which is one's subjective feeling that such influence exists. Perceptual agency increases when the outcome events in the game are both relevant to the decisions made in choice events, and contain consequences which are predictable at the decision point[3].

Example Player Agency Diagram



Research Questions

Does including player agency in the design of an educational game affect retention of educational content?

Does including player agency in the design of an educational game affect the play experience?

Methods

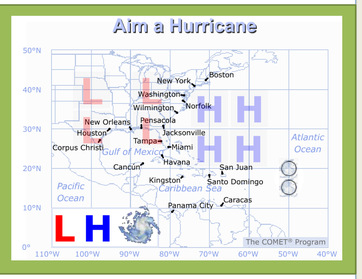
Design

The participants will be 120 eighth grade science students, randomized at the classroom level, from a suburban middle school in the greater Twin Cities area. The content of the games and the interactive visualization has been matched to that which is typically covered in the Earth Science unit on weather systems and will be administered as part of the normal unit activities.

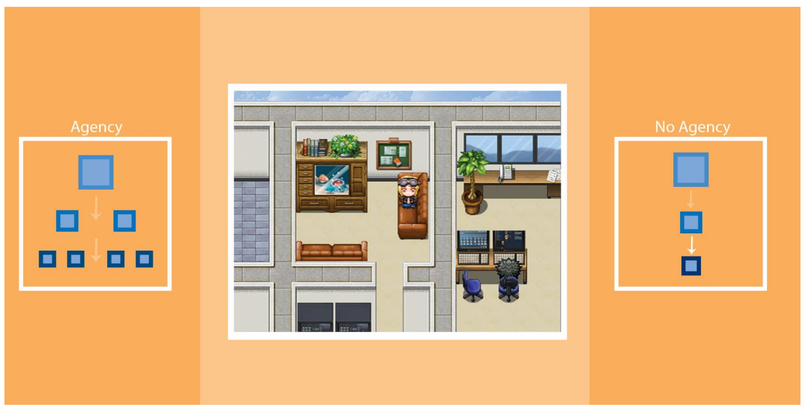
Interactive Visualizations (Control)



Interactive visualizations with game-like characteristics that are typical of the classroom teacher's normal activities in this unit, but which do not include agency and have relatively few game-like components (www.nhc.noaa.gov).



Game with Agency vs. Game without Agency



In both games, participants engage in data collection, use models, conduct interviews and report their findings. However, the sequence of narrative events is partially controlled by the player in the agency condition, whereas the condition without agency has a set path.

Measures

The Play Experience Scale

The Play Experience Scale (PES)[5] was developed as a means of assessing play experiences in video games in four categories:

Freedom

The player's sense of being able to do what they want to do within the game space	
If I wanted to do something in the game, I was able to do it	Freedom
I was able to make the game do what I wanted it to	Freedom
The game gave me the freedom to act how I wanted to	Freedom
The game made it difficult to preform the actions that I wanted to	Freedom (R)

No-extrinsic

The absence of extrinsic motivation, i.e.. a desire to please the researcher as the primary motivation while playing	
I was not worried about someone judging how I preformed in the game	No-extrinsic
Regardless of how I preformed in the game, I knew there wouldn't be a real-world consequence	No-extrinsic
My performance in the game was not going to matter outside the game	No-extrinsic
I felt like I had to do well, or the experimenter would judge me	No-extrinsic (R)

Autotelic-focus

The player's sense of immersion	
When I was using the game, I was focused on the task at hand	Autotelic-focus
I wanted to do well in the game, "just because"	Autotelic-focus
When I was using the game, I wanted to do as well as possible	Autotelic-focus
I tried to succeed in the game because I felt like it	Autotelic-focus

Play-direct

The 'gameness' of the game	
When I was using the game, I felt like I was playing rather than working	Play-direct
I would characterize my experience with the game as playing	Play-direct
I was playing a game rather than working	Play-direct
Using the game felt like work	Play-direct (R)

Content Assessment and Interviews

Additionally, we will be developing a multiple choice content assessment which will be administered both directly after the activities to assess what content was delivered and again one week later to help assess the retention of that content . We will also conduct interviews with selected participants about their experience with the games.

References

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2. Multi-Platform Gaming: For the Win! (2014) http://www.nielsen.com/us/en/insights/news/2014/multi-platform-gaming-for-the-win.html
3. Thue, D., Bulitko, V., Spetch, M., & Romanuik, T. (2010). Player Agency and the Relevance of Decisions. In R. Aylett, M. Y. Lim, S. Louchart, P. Petta, & M. Riedl (Eds.), *Interactive Storytelling* (pp. 210–215). Springer Berlin Heidelberg.
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5. Pavlas, D., Jentsch, F., Salas, E., Fiore, S. M., & Sims, V. (2012). The Play Experience Scale Development and Validation of a Measure of Play. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *54*(2), 214–225.